Celonis: From TUM Lab to €13 Billion

Teaching Case (HBS-style) — written for this course

Prepared for: TUM Entrepreneurship Course: Prototyping Entrepreneurial Ideas for AI

Decision Point: June 2016 — €10M ARR, Profitable, Growing 100% YoY. Raise or Bootstrap?

How this case ties into Module 12

You're deciding this week whether your company outlives the course; Celonis faced the grown-up version of the same decision. Work three connections: (1) **Idea 1 — raise or don't:** five profitable years BEFORE the first round is what 'default alive = leverage' looks like in practice — map their 2016 position onto the Raise-If/Don't-Raise slide. (2) **Idea 2 — the power law from the founder's side:** Accel needed a category-winner story; the founders had to decide whether to tell it. (3) **Idea 4 — your launchpad, literally:** Celonis started with €12,500 and an EXIST scholarship at TUM — the same program on your 'what happens Monday' slide. That's not a metaphor. Book the meeting.

Introduction

In 2011, three students at the Technical University of Munich (TUM) were doing IT consulting work alongside their studies when they ran into process mining — a technique for understanding how businesses actually run by analyzing operational data.

Process mining was not a sexy domain. It was academic. Boring. Enterprise software applied to operations and compliance.

Alexander Rinke, Bastian Nominacher, and Martin Klenk saw something different.

They saw that most companies had no idea how their business processes actually worked. They had process diagrams (how it should work) and they had systems (ERP, CRM) that recorded the actual work, but no one had connected those two things.

What if you took the data that operations systems already recorded and visualized how the business actually ran? What if you could see where time was being wasted, where rules were being broken, where processes were inefficient?

That would be software.

Five years later, in 2016, Celonis had: - Roughly €10 million in annual recurring revenue - Customers including Siemens and a roster of DAX-scale enterprises — plus SAP itself, which resold Celonis under its own price list - Organic growth (no venture capital — bootstrapped since day one) - A product that was genuinely solving real problems for large enterprises - A decision to make: **Scale it or keep it sustainable?**

This is the story of a European founder team that faced a high-class problem: success, and the leverage that came with it. They had built something customers wanted. Now they had to decide what kind of company they wanted to become.

Background

The Process Mining Insight (2010-2011)

Alexander Rinke, Bastian Nominacher, and Martin Klenk were students at TUM — Rinke in mathematics, Nominacher in finance and information management, Klenk in computer science. Consulting work alongside their studies (including a project untangling the IT service desk of Bayerischer Rundfunk) brought them into contact with process mining — a technique pioneered academically by Wil van der Aalst's group at TU Eindhoven, not at TUM. The founders were users of that research, not students of its inventor.

The context: - Enterprise software was fragmented: SAP for finance, Salesforce for sales, ServiceNow for IT, etc. - Each system had transaction logs (every action recorded with a timestamp) - But no one had aggregated these logs to understand the actual business process - When problems occurred (delayed orders, stuck approvals, customer complaints), teams didn't have a clear picture of where the breakdown happened

The insight: The three founders realized that the raw data to solve this problem already existed. Companies like SAP didn't need a new system; they needed a new lens on existing data.

If you took the data from an ERP system (order entry, fulfillment, invoicing, payment) and analyzed the sequences of events, you could reconstruct the actual business process. Not the intended process, but the real process with all its inefficiencies, workarounds, and exceptions.

This was powerful because: 1. **It was objective.** It wasn't based on what managers thought happened; it was based on what actually happened in the system. 2. **It revealed hidden waste.** A purchase order might be stuck in an unapproved state for 5 days because a

manager was on vacation. The process diagram said “1 day.” Reality said “5 days.” With process mining, you could see both. 3. **It enabled improvement.** Once you could see the real process, you could optimize it.

Building Celonis (2011-2015)

The three founders didn’t start as entrepreneurs. They started as students with a consulting project, trying to build better tools for seeing how organizations actually ran.

2011: They founded Celonis as a TUM spin-off — funded with roughly €12,500 of their own money plus the EXIST founder scholarship, the German federal grant for university spin-offs. Their first customer, Bayerischer Rundfunk, hired them to untangle its IT service desk.

2012-2013: They built out the product and showed it to companies with SAP systems. The response was strong: “This is incredibly useful. How much would you charge for this?” The company was profitable almost from the start — customer revenue funded everything.

2014-2015: Growth compounded. In 2015, SAP signed a reseller agreement: “SAP Process Mining by Celonis” went onto SAP’s own price list — a rare arrangement for a four-year-old startup.

The business model: - License software to enterprises with SAP, Oracle, or other ERP systems - Charge per implementation (€50k-200k+ per customer) - Plus annual maintenance/subscription - Target customers: Large enterprises (1000+ employees) with complex processes

The founding: - Three co-founders with technical credibility and a TUM spin-off pedigree - No external equity funding (no venture capital) - Started in Munich, in the heart of the German enterprise-software market - Bootstrapped on ~€12,500 of founder savings, the EXIST founder scholarship, and early customer revenue

The First Customers (2011-2015)

Celonis’s early customers were not random. They were:

1. **Customers who already knew the founders:** Relationships from their student consulting work and the TUM network
2. **Customers with SAP systems:** SAP’s ecosystem is where process mining adds most value
3. **Customers with acute pain:** Companies losing money due to inefficient processes, stuck orders, unnecessary manual work

Early wins:

Customer	Industry	Problem	Result
Large German Bank	Finance	Loan approval took 20 days; industry standard was 10 days	Celonis showed the bottleneck (missing documents). After fixing, 8 days.
Global Automotive Supplier	Manufacturing	Invoices stuck in approval queue. Missing revenue recognition.	Celonis revealed that approvers were never checking the queue. After process redesign, 95% auto-approved. €10M in cash flow improvement annually.
Bayerischer Rundfunk	Media	IT service-desk tickets took too long, and nobody could say where they got stuck	The founders' first-ever customer project: mining the ticket data revealed the bottlenecks — and produced a reference customer.

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

The pattern: Celonis didn't sell a product, they sold insight. Large enterprises would pay significant sums to understand where time and money were being wasted.

The Bootstrapped Model (2011-2016)

By early 2016, Celonis was: - **€10 million in ARR** (annual recurring revenue) - **Completely bootstrapped** (no VC funding) - **Highly profitable** (70-80% gross margins due to SaaS model; a lean team by enterprise-software standards) - **Growing 100% year-over-year** - **Located in Munich** (not Silicon Valley), giving them lower overhead

Financial snapshot (estimated):

Metric	Value	Notes
ARR	€10M	Recurring software revenue
Customers	50-75	Mostly large enterprises
Team size	50	30 engineers, 20 sales/operations
Office locations	Munich + 1 sales office	Minimal footprint
Burn/profit	Profitable	Likely 15-25% operating margin
Runway	Infinite	Generating more cash than spending

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The advantage of bootstrapping: - No investor pressure to "blitzscale" or hit arbitrary growth targets - No obligation to exit in 7-10 years - Profit was reinvested into hiring and product - Slower, but sustainable

The disadvantage of bootstrapping: - Limited capital to accelerate growth - Unable to invest heavily in sales/marketing - Competitors with funding could move faster - Couldn't afford expensive enterprise sales infrastructure (multiple sales regions, large pre-sales teams)

The Competitive Landscape (2015-2016)

By 2016, process mining was beginning to get attention:

Competitor	Approach	Funding	Strength
Celonis	Self-serve + sales	Bootstrapped	Best UX, founders understand the problem
Fluxicon Disco	Lightweight desktop tool from a TU Eindhoven spin-off	Bootstrapped	Good algorithms, loved by analysts — not built for enterprise scale
Forrester reports mentioning process mining	Various enterprises starting to notice	Various	Growing market awareness
Larger analytics companies	Might add process mining as a module	VCs funding analytics broadly	Large incumbent threat

The market was small but growing. The founders knew that if process mining became a category, a large software company (SAP, Oracle, Salesforce, Workday) would either: 1. Build their own solution 2. Acquire a leader like Celonis

The window to establish market leadership was open but closing.

The Situation

June 2016: The Decision Moment

By June 2016, Celonis was at an inflection point. Multiple pressures converged:

Pressure 1: Investor Interest

Word of mouth had spread in the venture capital world. Multiple VCs (including prominent Silicon Valley firms) were approaching the founders:

> "We know you're bootstrapped and profitable. But we could accelerate your growth significantly. With funding, you could expand to the US market, hire a larger sales team, and establish market leadership before competitors move in."

Some offers were specific: - **Series A (€15-25M):** Build US presence, expand product, grow to 500+ customers - **Larger round (€50M+):** Blitzscale globally, hire 300+ employees in 18 months, become the "Celonis of the world"

Pressure 2: Market Opportunity

Process mining was becoming a recognized category. Gartner added it to their radar. McKinsey published reports on operational efficiency. SAP was already reselling Celonis under its own brand.

The founders knew: whoever established process mining as a software category would own a large market. Current size: €500M globally (estimated). Projected size in 10 years: €5B+.

Celonis had a 2-3 year window to establish itself as the leader. After that, larger companies would copy them.

Pressure 3: Competitive Pressure

Software AG's ARIS unit already sold process-performance tooling, and giants like SAP or IBM could build — or acquire — their way in at any time. If one of them moved, Celonis's window would close. They had to move fast.

Pressure 4: The Personal Factor

The three founders (now in their late 20s / early 30s) were at a life stage where they were thinking about: - Building a “unicorn” (billion-dollar company) - Achieving massive impact - Proving themselves as entrepreneurs, not just researchers

The vision of a €10M annual revenue company (impressive by German standards) paled compared to the vision of a €1B+ company.

The Options Before Them

The founders had to decide: **Remain bootstrapped and scale sustainably, or raise capital and pursue aggressive growth?**

Option A: Remain Bootstrapped and Scale Sustainably

The thesis: - Celonis is profitable and growing 100% YoY - Profitability means we don’t need external capital - Slow growth is still good growth - We maintain complete control

The plan: - Continue to grow organically - Hire 20-30 people per year (not 100+) - Expand to 1-2 new markets (UK, Benelux) without full blitzscale - Build sustainable unit economics - Exit in 5-10 years with a €1-2B valuation (reasonable for a €100-200M ARR company)

Pros: - Founders maintain control and equity - Company remains profitable and sustainable - No pressure to hit unrealistic growth targets - Can focus on product quality vs. growth at all costs - Lower risk (not dependent on market conditions or investor sentiment)

Cons: - Competitors might move faster and establish market position - SAP/IBM might enter the space and crush Celonis - Misses the “blitzscale” opportunity to become a multi-billion-euro company - Slower path to wealth for the founders - Team might leave to work at better-funded competitors

Option B: Raise a Series A (€15-25M) and Pursue Moderate Growth

The thesis: - Use capital to accelerate without losing control - Expand to US market (where enterprise software is bought at scale) - Hire a larger sales team to penetrate existing customer base - Still maintain founder control and DNA

The plan: - Raise Series A at favorable terms (founders retain 70%+ of company) - Use capital to hire 80-100 people (mostly sales and customer success) - Expand from 50 to 200+ customers in 12-18 months - Establish offices in Silicon Valley and NYC for US sales - Maintain engineering quality and product vision

Pros: - Accelerate growth without losing founder control - Access to better capital and relationships - US market is where the real money is in enterprise software - Can build sustainable sales model before competitors move in - Still path to a €5-10B valuation

Cons: - Founder dilution (probably 20-30%) - Now accountable to investors (not just customers) - Pressure to hit growth targets (if growth slows, investors will push harder) - Sales organization might change the culture - €15-25M will feel small once spent; might face pressure to raise Series B quickly

Option C: Raise a Large Round (€50M+) and Blitzscale

The thesis: - Go big or go home - The software market rewards scale (network effects, switching costs, market share) - Blitzscale now, dominate the market, and become the "Celonis of the world" - If we don't, a funded competitor will

The plan: - Raise a large Series A from top-tier VCs (€50-100M) - Hire 200-300 people in 12 months - Expand aggressively to US, Asia, Europe - Build a full enterprise sales organization - Invest heavily in product development and brand - Aim for €1B valuation in 5 years, unicorn status

Pros: - Dominate market before competitors move in - VCs provide not just capital but network and credibility - Becomes a venture-scale company with massive growth potential - Founders become billionaires (on paper) if successful - Founders can attract top talent (equity upside matters)

Cons: - Massive dilution (founders might own 30-40% of company after 2-3 rounds) - Loss of control (VCs will demand board seats, decision-making power) - Pressure to hit hockey-stick growth targets - Culture changes as you grow 300+ people in 12 months - High risk: if growth stalls, company implodes (profitability doesn't matter to VCs) - Founders might be replaced if growth slows (common VC playbook)

The European Context

Celonis was building in Germany, not Silicon Valley. This created specific strategic considerations:

The German SaaS context (2016): - German software companies typically bootstrapped or took smaller raises - Growing companies had two paths: sell to SAP/IBM or slowly become €100M ARR company - Blitzscaling was not the German default

The SAP factor: - SAP's ecosystem was where Celonis lived - SAP was already a partner: since 2015 it had resold Celonis through its own price list — and partners can become acquirers, or competitors - If SAP saw Celonis as strategic, they might acquire (likely €500M-1B+) - If SAP saw Celonis as a threat, they might build a competing product

The US market factor: - European software rarely succeeded in the US without US capital and US team - To compete in the US, Celonis would need US presence, US board, US sales - Raising from US VCs was the most credible way to establish US presence

What We Know About Process Mining's Future (Spoiler-Free)

We can't tell the founders what actually happens. But we can note: - By 2020, process mining becomes a recognized software category - Multiple heavyweights buy their way in (UiPath acquires ProcessGold in 2019, SAP acquires Signavio in 2021, Microsoft acquires Minit in 2022) - Celonis remains the leader - The founders' actual decision is one of the options described above — the epilogue, to be read after class, tells you which

But in June 2016, none of this was known.

The Decision Point

It is **June 2016.** Alexander Rinke, Bastian Nominacher, and Martin Klenk sit together in their Munich office.

On the table: - A Series A term sheet: **$27.5M, co-led by Accel and 83North** — a storied Silicon Valley firm and a top London/Tel Aviv fund - A larger offer from a growth equity firm (€50M, more aggressive terms) - Their own financial model showing they can profitably grow to €50M ARR in 5-7 years without external funding - An open question about SAP: deepen the reseller partnership, explore an acquisition conversation, or brace for competition?

The founders must decide:

The Core Question

Do Celonis raise capital to aggressively scale and dominate the process mining market globally, or do they remain bootstrapped and build a sustainable, profitable company that might eventually be acquired or go public?

And underneath: **What kind of founders do you want to be?**

The Sub-Questions

1. **Market timing:** Is the window to establish market leadership going to close? If Celonis doesn't raise and scale now, will competitors beat them? Or is the market large enough that multiple players can succeed?

2. **Founder control:** How much equity would the founders be willing to give up to accelerate the company? At what point does the company stop being "theirs"?
3. **The SAP factor:** Should the founders explore a partnership or acquisition with SAP (their largest customer and potential acquirer)? Or does that seem like selling out before the company reaches its potential?
4. **The personal trade-off:** Scaling to a multi-billion-euro company requires 5-7 years of very hard work, possibly replaced as CEO if you're not the right leader, massive pressure, and risk. Is that worth it? Or is a profitable, sustainable €100M ARR company with full founder control actually better?
5. **The definition of success:** What does success actually mean to these founders? Wealth? Impact? Control? Freedom? The answer determines which path they choose.

Exhibits

Exhibit A: SAP and Process Mining (2016 context)

SAP was beginning to recognize process mining as strategic:

Year	SAP Action
2015	Signed a reseller agreement with Celonis: "SAP Process Mining by Celonis" sold through SAP's own price list
2015-2016	Co-sold Celonis into its enterprise installed base; process mining visibly on SAP's strategic radar
Later (2021)	Acquired Signavio — a Celonis competitor in process intelligence — for a reported $1B+, turning the former partners into rivals
Never	SAP never acquired Celonis. Celonis remains an independent company to this day

Context: SAP recognized that process mining could become strategic to their enterprise software platform. They could build, buy, or partner — and in 2016 they were partnering, with Celonis. What the founders had to weigh was how long that would last, and what SAP's next move would be.

Exhibit B: Growth Projections Under Different Scenarios

Scenario A: Remain Bootstrapped

Year	ARR	Team	Founder Ownership
2016	€10M	50	100%
2017	€20M	75	100%
2018	€40M	120	100%
2019	€80M	200	100%
2020	€150M	350	100%

(Slower growth, but complete ownership)

Scenario B: Series A (≈€20-25M)

Year	ARR	Team	Founder Ownership	Valuation
2016	€10M	50	80%	€100M
2017	€30M	150	75%	€300M
2018	€60M	300	70%	€600M
2019	€100M+	400	65%	€1B
2020	€200M	500	60%	€2B+

(Faster growth, some dilution)

Scenario C: Large Round (€50M+, Blitzscale)

Year	ARR	Team	Founder Ownership	Valuation
2016	€10M	50	40%	€250M
2017	€50M	300	35%	€500M
2018	€100M	500	30%	€1B
2019	€200M	800	25%	€2B
2020	€300M+	1000+	20%	€3B+

(Maximum growth, significant dilution, unicorn potential)

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates. Actual results depend on market conditions, execution, and luck.

Exhibit C: Founder Wealth Scenarios

Under Scenario A (Bootstrapped): - 2020 valuation: €1.5B (estimated) - Founder ownership: 100% - Per-founder value: €500M (if sold equally) - Timeline: 7-10 years

Under Scenario B (Series A): - 2020 valuation: €2.5B - Founder ownership: 60% - Per-founder value: €500M - Timeline: 5-7 years

Under Scenario C (Blitzscale): - 2020 valuation: €3B+ - Founder ownership: 20% - Per-founder value: €200-300M per founder - Timeline: 4-5 years - Risk: If company fails or stalls, founders might be replaced

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit D: Decision Matrix

Factor	Bootstrap	Series A	Blitzscale
Control	High	Medium	Low
Speed to market	Slow	Medium	Fast
Risk	Low	Medium	High
Potential upside	€1-2B	€2-5B	€5-10B+
Founder wealth	€300-500M	€300-500M	€200-400M (if successful)
Founder role	Maintains control	Shares control	Might be replaced
Timeline to exit	8-10 years	6-8 years	5-7 years
Probability of success	Medium (70-80%)	Medium (60-70%)	Medium (40-50%)

Note: “Success” = reaching €1B+ valuation. The bootstrap path has highest probability but slower timeline. Blitzscale has highest upside but highest risk.

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit E: The Term Sheet (Series A Option)

A reconstruction of the term sheet on the table. (The real round, announced June 2016, was a $27.5M Series A led by Accel and 83North; its valuation was never publicly disclosed.)

Term	Value
Investment amount	$27.5M (≈€24M)
Pre-money valuation	≈€100M
Post-money valuation	≈€125M
Investor ownership	~20%
Founder dilution	~20% (founders still own ~80%)
Board composition	5 seats: 2 founder, 2 investor, 1 independent
Investor rights	Information rights, registration rights, pro-rata rights
Vesting schedule	4-year vesting for founder equity (standard for new hires, not founders)
Liquidation preference	1x, non-participating (standard, not aggressive)

What this means: - Founders would own ~80% of a ≈€125M company — a stake worth ≈€100M on paper, roughly €33M per founder if split equally - Investors get 2 board seats and have information rights - Founders still have significant control (majority of board) - Relatively founder-friendly terms

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit F: The SAP Acquisition Path (Alternative)

SAP's closeness to Celonis raised another theoretical option: **early acquisition.**

An acquisition by SAP might look like:

Scenario	Valuation	Founder Wealth	Timing	Outcome
Early (2016)	€300-500M	€100-150M per founder	Immediate	Founders become employees/ board members
Mid (2018)	€1-2B	€300-600M per founder	2 years	Founders lead Celonis division at SAP
Late (2020)	€3-5B	€900M-1.5B per founder	4 years	Founders negotiate exit

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates. (Spoiler guard: this exhibit models a hypothetical path, not history.)

The question: **Is a 2-year path to €500M of wealth, with reduced control, better than a 7-year path to €1-2B of wealth with full control?**

Different founders answer this differently.

Questions for Class Discussion

1. **The market timing question:** How do you know if you have a 2-year window to establish market leadership, or a 10-year window? What signals would tell the founders whether they need to rush?
2. **The founder role question:** As a founder, would you prefer to own 100% of a €100M company (still significant) or 30% of a €3B company (much wealthier on paper, but less control)? How do you make this choice?
3. **The SAP question:** If your most important partner is also your potential acquirer — or your future competitor — how do you navigate that relationship? Do you: - Deepen the partnership (might strengthen your distribution — and your dependence)? - Avoid conversations with them (protect independence)? - Open acquisition discussions proactively (reduce uncertainty)?
4. **The European context:** Would you make a different decision if Celonis was located in Silicon Valley? Does location matter?
5. **The founder fit question:** Do the three founders have the skills to scale a multi-billion-euro company, or are they better suited to building a sustainable €100M company? Does this factor in?

6. **The irreversibility question:** Once you raise VC money, it's hard to go back to bootstrapped. But if you stay bootstrapped, you can always raise later. Does the option value of waiting matter?
7. **The employee perspective:** If you were an engineer at Celonis, would you prefer: - Bootstrapped path (lower compensation risk, more stable, slower wealth building)? - Series A path (better compensation, some wealth via equity upside, growth)? - Blitzscale path (higher risk, massive equity potential, high-pressure growth)?

Epilogue (read after class)

What actually happened:

They raised. In June 2016, after five profitable, bootstrapped years, Celonis closed a **$27.5M Series A led by Accel and 83North** — the first institutional money since starting in 2011 with roughly €12,500 of pooled savings and an EXIST founder scholarship.

Then it compounded. In 2018, Accel and 83North doubled down with a **$50M Series B at a $1 billion valuation** — unicorn status. A $290M Series C at $2.5 billion followed in 2019. In 2021 came a **$1B Series D at roughly $11 billion**, and in 2022 a **$400M Series D expansion led by the Qatar Investment Authority at roughly $13 billion** — making Celonis one of the most valuable software companies Europe has ever produced.

What did not happen: SAP never acquired Celonis. The reseller partnership eventually gave way to rivalry — SAP bought Celonis competitor Signavio in 2021 — and Celonis stayed independent, with the founders still holding the levers. Because they raised from strength rather than need, they kept the control that desperate founders sign away.

The teaching point: Celonis raised when they didn't have to. Five years of default-alive profitability is precisely what made the terms good — investors compete hardest for companies that can walk away. The lesson is neither "bootstrap forever" nor "raise fast." It is: build the option, then exercise it deliberately. The bootstrap-vs.-raise decision is not binary; the key is making the choice intentionally, with a clear view of the trade-offs.

Full circle: the ~$13 billion company started with founder savings and an EXIST founder scholarship — the same federal grant we will point you toward after demo day. Worth remembering as you make your own continuation decision.

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